SQL Aggregate Functions & Operations
Course: BrightLearn Basic SQL
Exercise 2

01 The Students Table

(01)
```
SELECT DISTINCT department
FROM students;
```

department
IT
HR
Finance

(02)
```
SELECT department,
AVG(age) AS avg_age
FROM students
GROUP BY department;
```

department	avg_age
IT	20.5
HR	22
Finance	23

(03)
```
SELECT department,
COUNT(*) AS student_count
FROM students
GROUP BY department
HAVING COUNT(*) > 1;
```

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department	student_count
IT	2
HR	2

(04)
```
SELECT   student_id,
           name,
            age,
         department
FROM     students
WHERE    age BETWEEN 21 AND 23;
```

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

(05)
```
SELECT    student_id,
            name,
             age,
          department
FROM      students
WHERE     department  IN ('IT', 'HR')
 AND       age > 21;
```

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

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02 The Courses Table

(06)
```
SELECT department,
SUM(credits) As total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;
```

department	total_credits
IT	11

(07)
```
SELECT course_id,
       course_name,
       department,
       credits
FROM courses
WHERE credits != 4;
```

course_id	Course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

(08)
```
SELECT course_id,
       course_name,
       credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

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course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

03 The enrollments Table

(09)
```
SELECT  MAX(grade) As max_grade,
        MIN(grade) As min_grade,
        AVG(grade) As avg_grade
FROM    enrollments;
```

max_grade	min_grade	avg_grade
90	78	84.6

(10)
```
SELECT  course_id,
        COUNT(enrollment_id) As enrollment_count
FROM    enrollments
GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

04 The salaries Table

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(11)
```
SELECT department,
SUM(salary) As total_salary,
SUM(bonus) As total_bonus
FROM salaries
GROUP BY department;
```

department	total_salary	total_bonus
IT	122 000	10 600
HR	109 000	7 500
Finance	70 000	6000

(12)
```
SELECT department,
AVG(salary) As avg-salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;
```

department	avg-salary
IT	61000
Finance	70000

(13)
```
SELECT employee_id,
       name,
       salary,
       bonus,
salary + bonus As total_compensation
FROM salaries
WHERE salary + bonus > 60000;
```

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employee_id	name	department	salary	bonus	total_compensation
1	Tom	IT	60 000	5000	65 000
3	Spike	Finance	70 000	6000	76 000
4	Tyke	IT	62 000	5600	67 600

05 The projects Table

(14)
```
SELECT department,
SUM (budget) As total_budget,
AVG (budget) As avg_budget
FROM projects
GROUP BY department
HAVING AVG (budget) > 70000;
```

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

(15)
```
SELECT project_id,
       project_name,
       department,
       budget
FROM projects
WHERE budget BETWEEN 50000 AND 120 000
 AND department <> 'Marketing';
```

project_id	project_name	department	budget
1	AI app	IT	120 000
2	Payroll System	Finance	80 000
3	Dashboard	IT	150 000
5	HR Portal	HR	50 000

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